

CUSTOMER SEGMENTATION & CAMPAIGN RESPONSE PREDICTION

Marketing Analytics Report

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Dataset: Customer Personality Analysis (Kaggle)

2,058 Total Customers	15.2% Response Rate	40.7% Best Segment	227% ROI Lift
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Executive Summary

This report presents a comprehensive analysis of customer data to support targeted marketing campaign decisions. Using advanced clustering and machine learning techniques, customers were segmented into actionable groups, and a predictive model was developed to identify likely campaign responders.

Metric	Value
Total Customers Analyzed	2,058
Overall Response Rate	15.2%
Best Performing Segment	Ultra Premium (40.7% response)
Best Prediction Model	Random Forest (F1-Score: 0.558)
ROI Improvement	227% lift by targeting top 20%

Recommendation: Target Ultra Premium and Premium Spenders first for maximum ROI.

1. Introduction

Businesses spend significant resources on marketing campaigns, but not all customers respond equally. This project was designed to optimize campaign spend by addressing three core objectives:

- Segment customers into meaningful groups using clustering algorithms
- Build a prediction model to identify likely campaign responders
- Provide actionable recommendations for campaign optimization and ROI improvement

2. Data Overview

Aspect	Details
Source	Customer Personality Analysis Dataset (Kaggle)
Original Records	2,240
Final Records (after cleaning)	2,058
Features	29 original + 15 engineered = 44 total
Target Variable	Response (1 = accepted campaign, 0 = did not accept)

Response Distribution

- Responded: 313 customers (15.2%)
- Did Not Respond: 1,745 customers (84.8%)

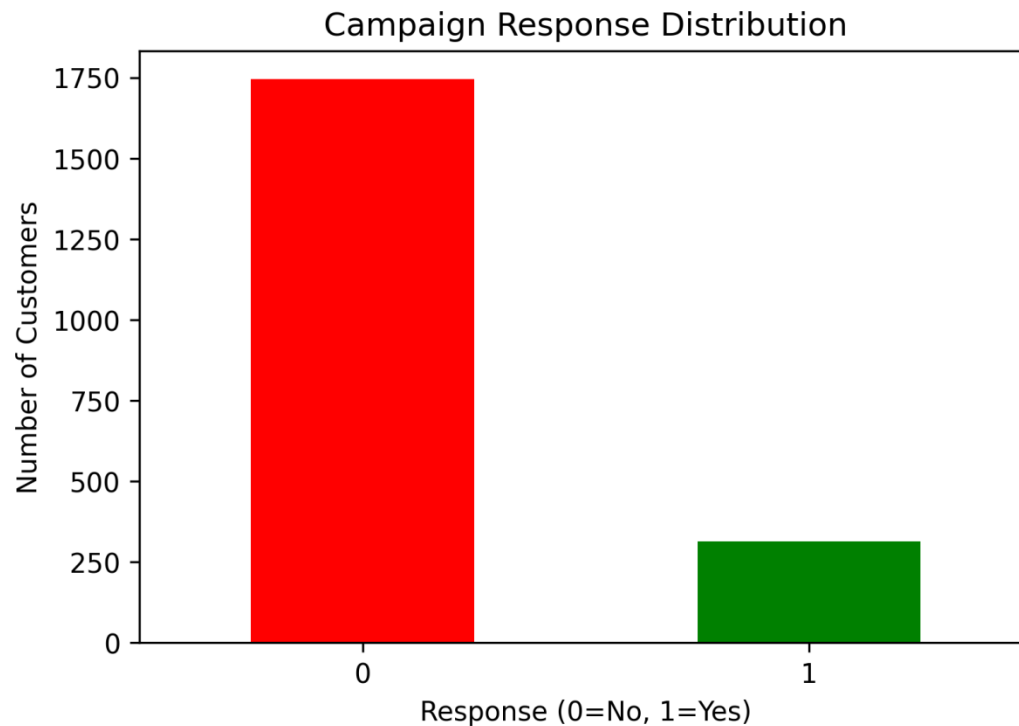


Figure 1: Campaign Response Distribution - Only 15.2% of customers responded

Insight: Imbalanced dataset — required SMOTE (Synthetic Minority Oversampling Technique) for effective modeling.

3. Data Preprocessing

3.1 Missing Values

The Income column had 24 missing values (1.07%). These were filled with the median income to preserve data integrity and avoid bias from row removal.

3.2 Columns Removed

Column	Reason for Removal
ID	100% unique values — would cause model overfitting
Z_CostContact	Constant value (3) — carries no information
Z_Revenue	Constant value (11) — carries no information

3.3 Outlier Treatment

Feature	Issue	Action Taken
Age	Values >100 years (customers born 1893)	Capped at 100 years
Income	Values up to \$666,666	Capped at 99th percentile (\$94,545)

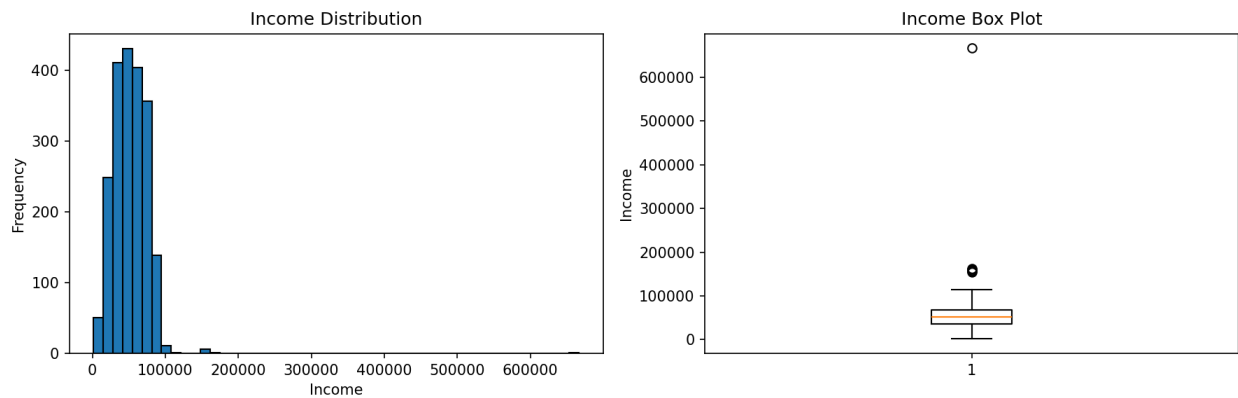


Figure 2: Income Distribution - Right-skewed with outliers capped at 99th percentile

3.4 Feature Engineering

15 new features were created across four categories to enhance model predictive power:

Category	Features Created
Spending Features	Total_Spending, Spending_Per_Capita
Purchase Features	Total_Purchases, Online_Purchase_Ratio
Campaign Features	Campaign_Success_Rate, Total_Campaigns_Accepted
Value Features	CLV_Score, Income_Per_Capita

3.5 Some Feature Graphs

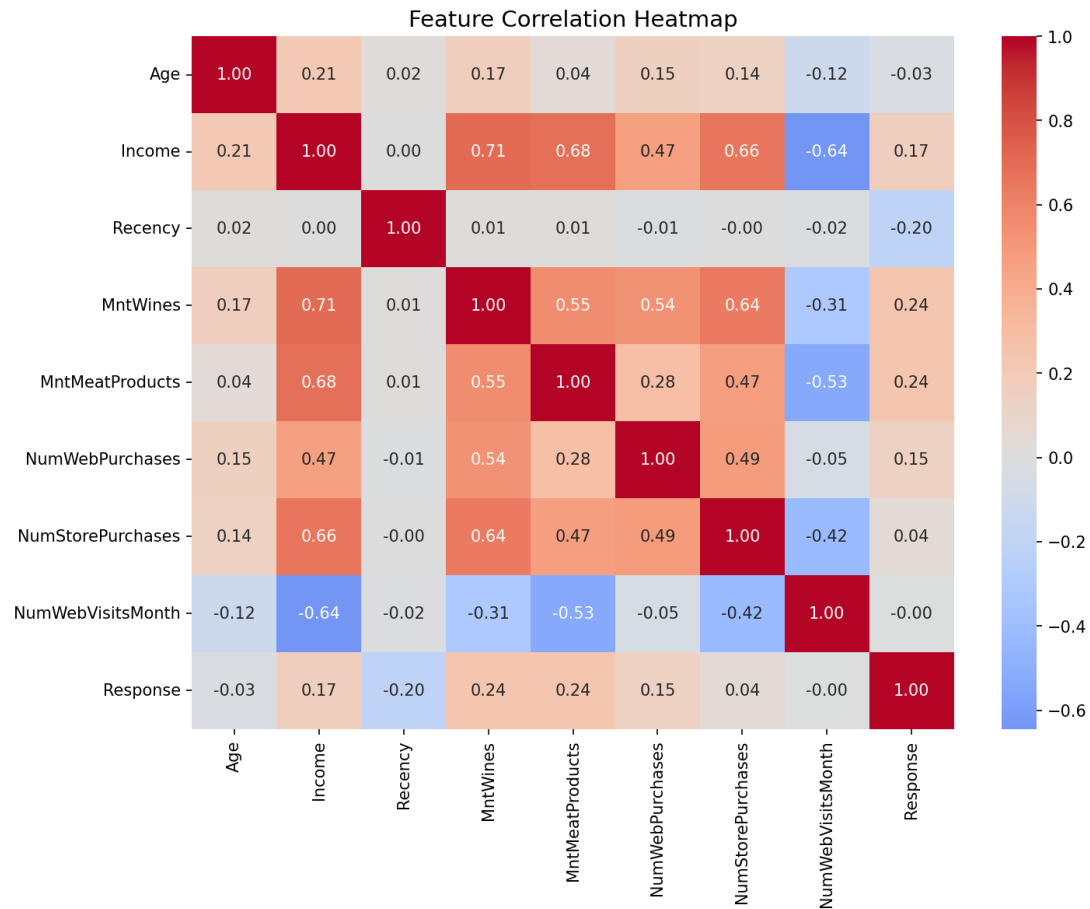


Figure 3: Correlation heatmap showing relationships between features and Response

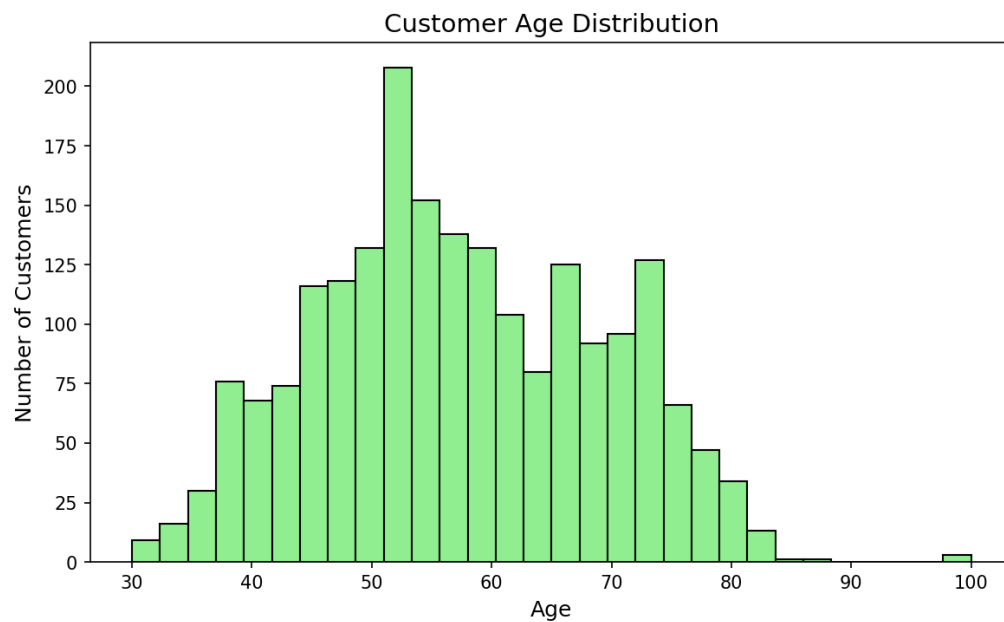


Figure 4: Age distribution of customers (capped at 18-100 years)

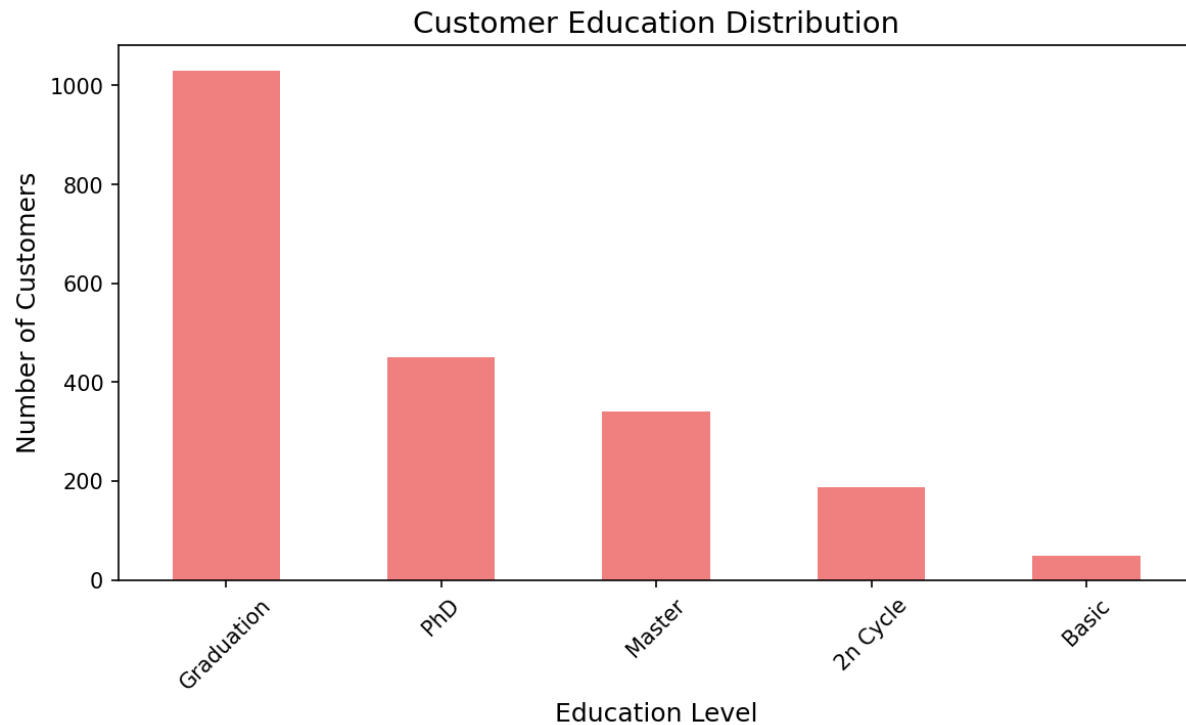


Figure 5: Education level distribution after ordinal encoding

4. Customer Segmentation (Clustering)

4.1 Feature Selection Experiment

Four feature combinations were tested to identify the optimal clustering inputs, evaluated using the Silhouette Score:

Combination	Features Used	Silhouette Score
Economic Focus ✓	Income, Total_Spending, Spending_Per_Capita	0.440
Demographic Focus	Age, Income, Education_Level	0.309
Full Combination	7 features	0.263
Behavior Focus	Online_Ratio, Purchases, Campaign_Success	0.248

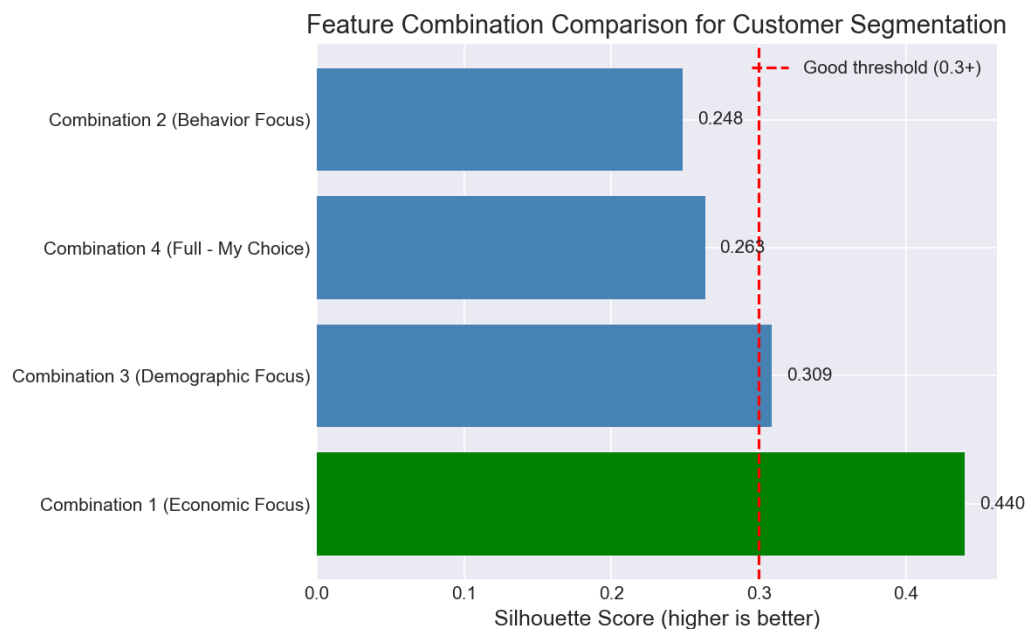


Figure 6: Silhouette Score comparison across feature combinations

Winner: Economic Focus combination (Silhouette Score: 0.440)

4.2 Optimal Number of Clusters

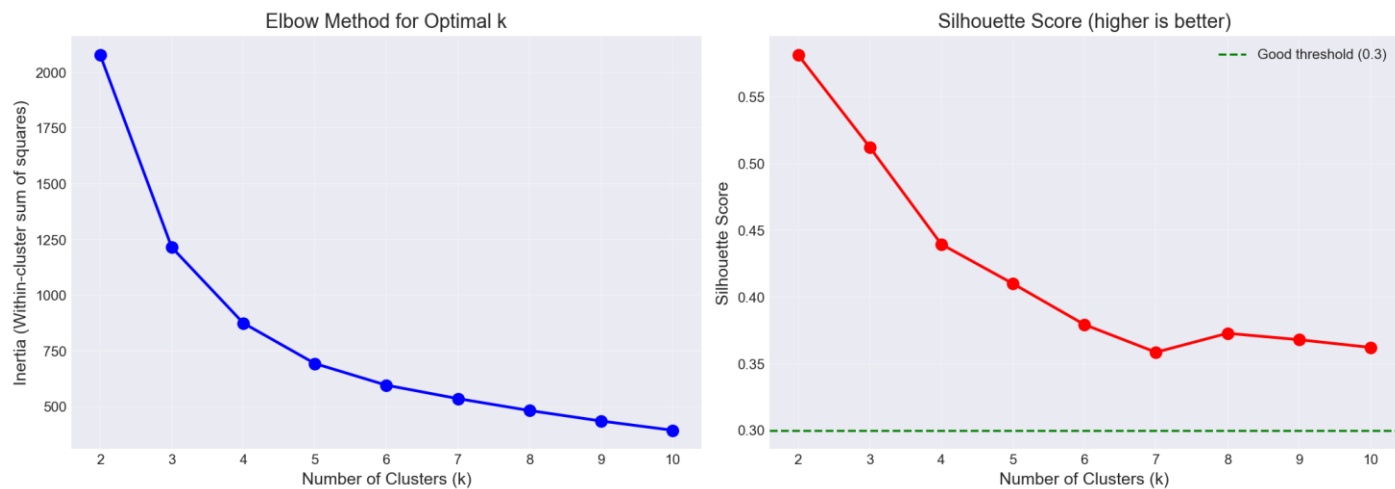


Figure 7: Elbow Method and Silhouette Score for k=2 to k=10

Clusters (k)	Silhouette Score	Business Value
k=2	0.582 (mathematical best)	Too broad — only rich vs poor distinction
k=4 ✓	0.440	4 actionable business segments

Decision: Selected k=4 for business interpretability over mathematical optimality.

4.3 Customer Segments

Segment	Size	Avg Income	Avg Spending	Response Rate
Ultra Premium	263 (12.8%)	\$79,053	\$1,706	40.7%
Premium Spenders	474 (23.0%)	\$69,609	\$1,087	15.2%
Budget Conscious	736 (35.8%)	\$29,544	\$87	11.0%
Middle Class	585 (28.4%)	\$53,190	\$376	9.1%



Figure 8: Customer segments visualized using PCA (88.44% variance explained)

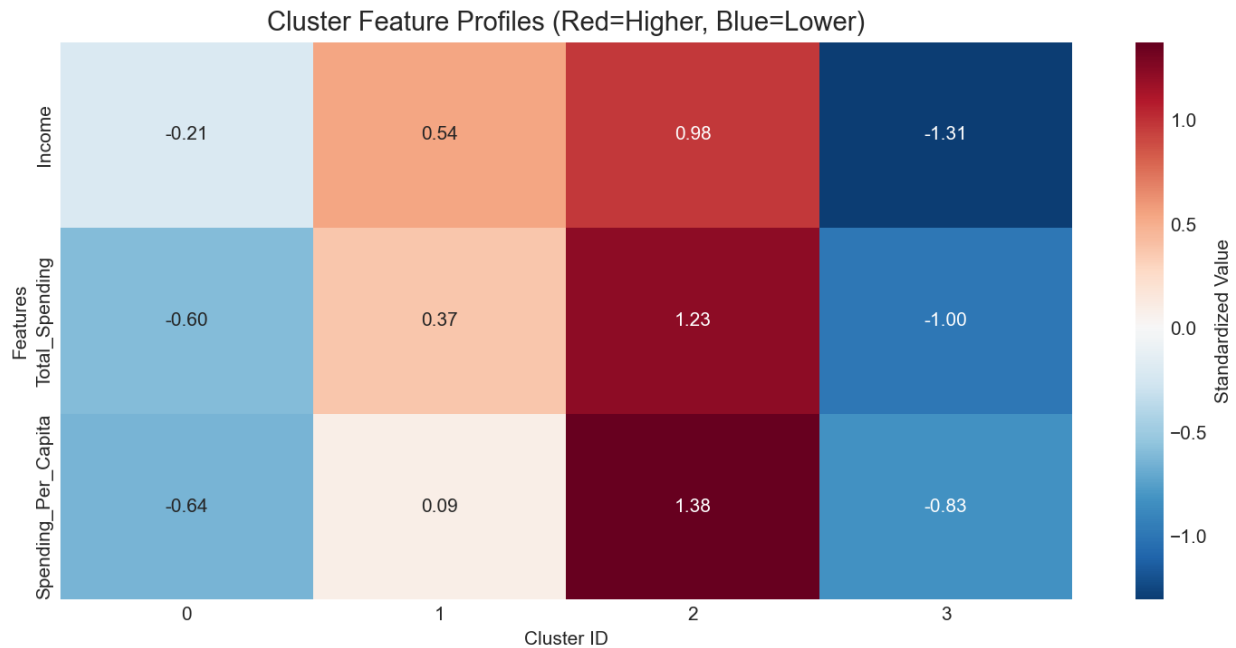


Figure 9: Cluster feature profiles (Red=Higher, Blue=Lower)

4.4 Key Segmentation Insights

Ultra Premium (12.8% of customers)

- Highest income and spending across all segments
- 40.7% response rate — 2.7x the overall average
- Recommended Action: VIP treatment, exclusive offers, early access programs

Premium Spenders (23.0% of customers)

- High income and high spending behavior
- 15.2% response rate, matching the overall average
- Recommended Action: Standard campaign inclusion, loyalty rewards

Budget Conscious (35.8% of customers)

- Low income, minimal spending patterns
- 11.0% response rate
- Recommended Action: Discount-driven emails, value-based messaging only

Middle Class (28.4% of customers)

- Moderate income and spending

- 9.1% response rate — lowest of all segments
- Recommended Action: Build engagement first via newsletters and brand-building content

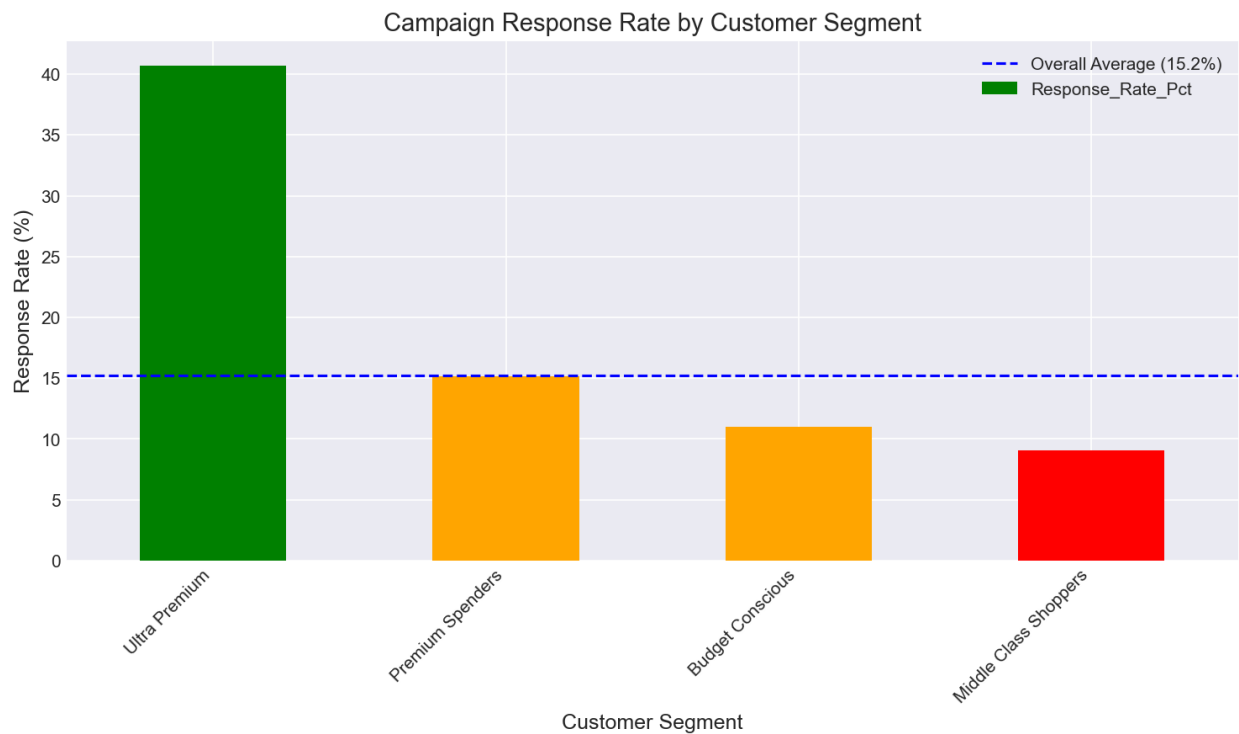


Figure 10: Campaign response rate by customer segment

5. Prediction Models

5.1 Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest ✓	84.7%	50.0%	68.3%	55.8%	87.9%
Logistic Regression	82.8%	45.6%	65.1%	53.6%	82.3%
Decision Tree	84.7%	50.0%	47.6%	48.8%	83.5%

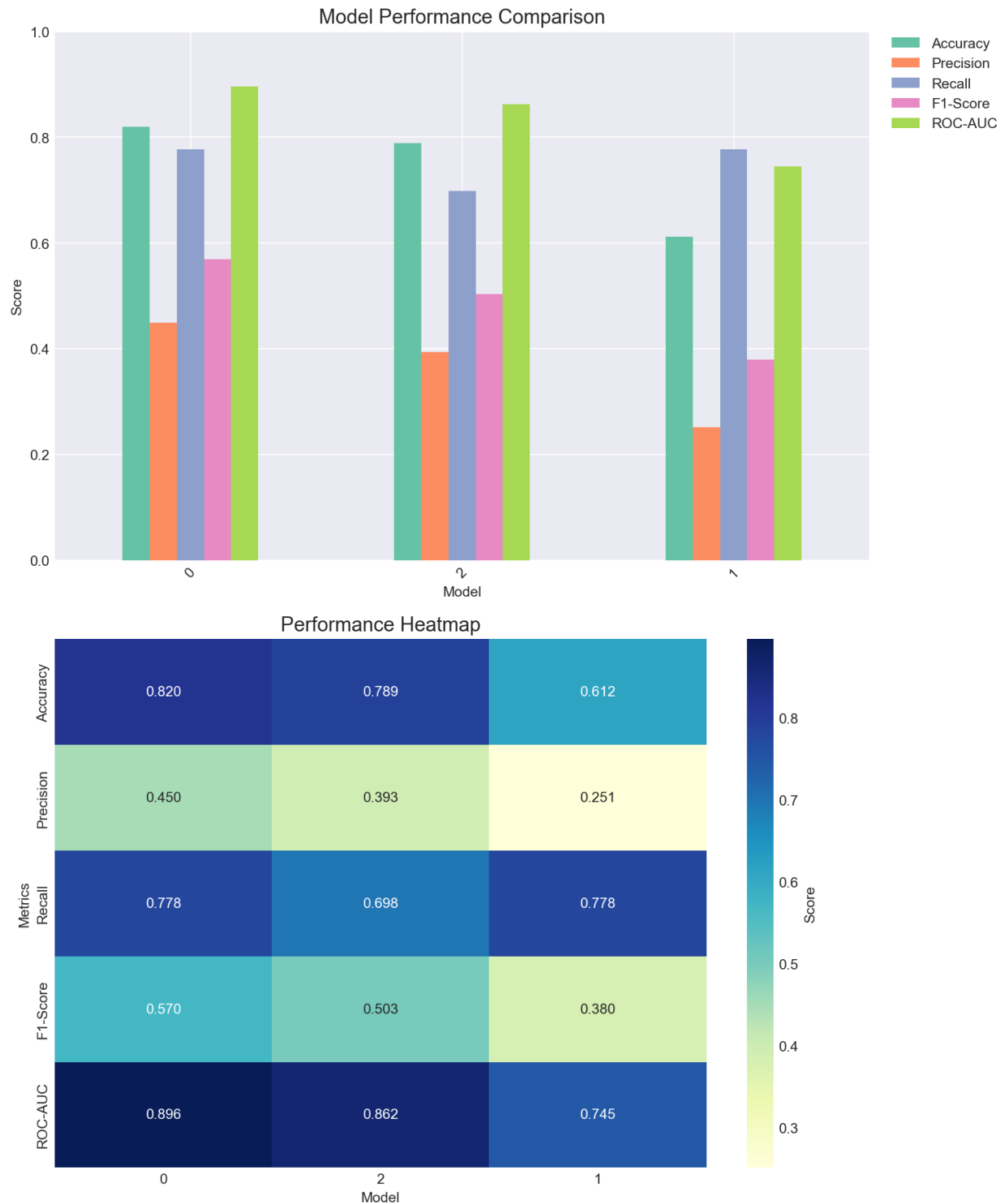


Figure 11: Model performance comparison - Random Forest performs best

Winner: Random Forest — Highest F1-Score (55.8%) and ROC-AUC (87.9%)

5.2 Why F1-Score Over Accuracy?

With imbalanced data (only 15.2% response rate), raw accuracy is a misleading metric. A model predicting 'No' for every customer would achieve 84.8% accuracy but zero practical value. The F1-Score balances Precision (how often predictions are correct) and Recall (how many actual responders are captured), making it the appropriate evaluation metric.

5.3 Confusion Matrix — Random Forest

	Predicted NO	Predicted YES
Actual NO	301 (73.1%) — True Negative	48 (11.7%) — False Positive
Actual YES	20 (4.9%) — False Negative	43 (10.4%) — True Positive

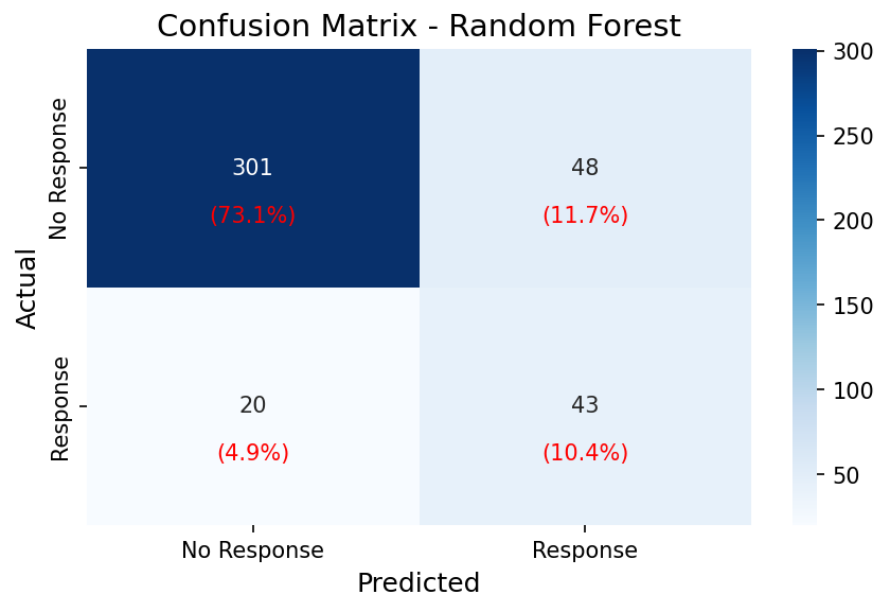


Figure 12: Confusion Matrix - Random Forest (43 correct YES predictions)

- True Positives: 43 correct 'YES' predictions
- False Positives: 48 customers incorrectly predicted as responders
- False Negatives: 20 actual responders missed by the model
- True Negatives: 301 correct 'NO' predictions

5.4 Feature Importance

Feature	Importance	Business Meaning
Campaign_Success_Rate	45.4%	Past responders are most likely to respond again
Recency	15.7%	Recent purchasers respond less to campaigns
Income_Per_Capita	11.6%	Higher income correlates with higher response
Total_Spending	9.4%	Higher spending predicts higher response
MntMeatProducts	8.3%	Meat buyers represent engaged customers

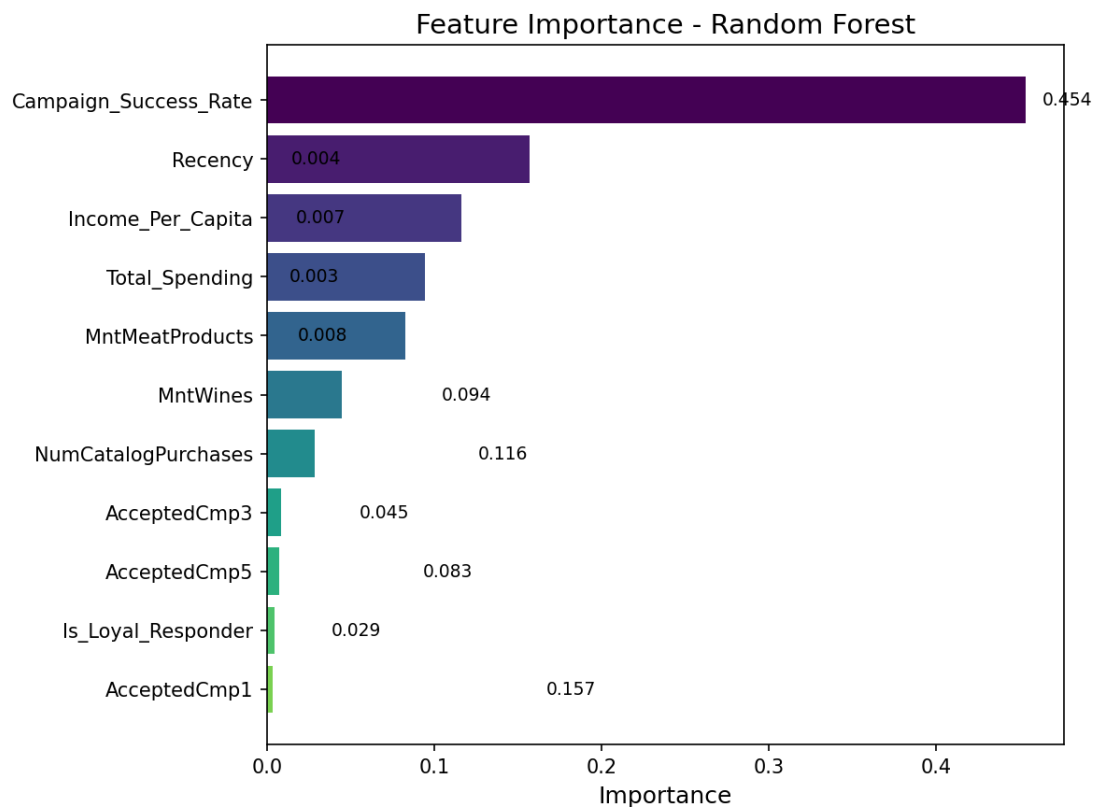


Figure 13: Feature importance - Campaign_Success_Rate dominates at 45.4%

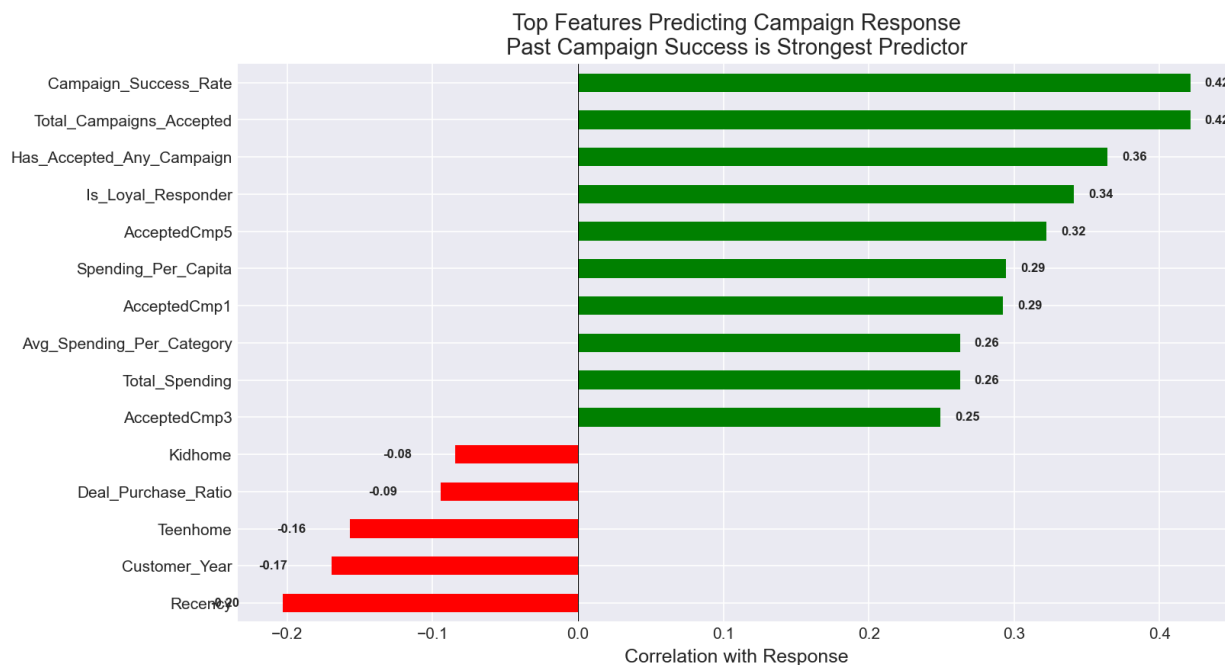


Figure 14: Feature correlations with campaign response

5.5 ROC Curves

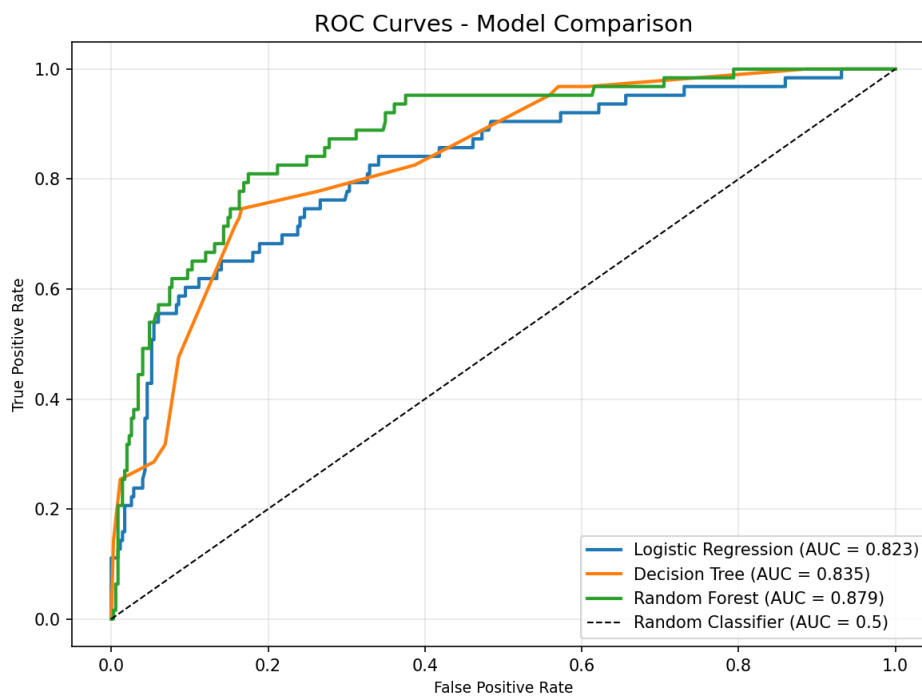


Figure 15: ROC Curves - Random Forest achieves AUC of 0.879

5.6 Cross-Validation Results

Metric	Value
Mean CV F1-Score	0.820
Standard Deviation	0.021
Test F1-Score	0.558

5.7 Hyperparameter Tuning

Parameter	Default	Best Tuned
n_estimators	100	200
max_depth	5	10
min_samples_split	2	2

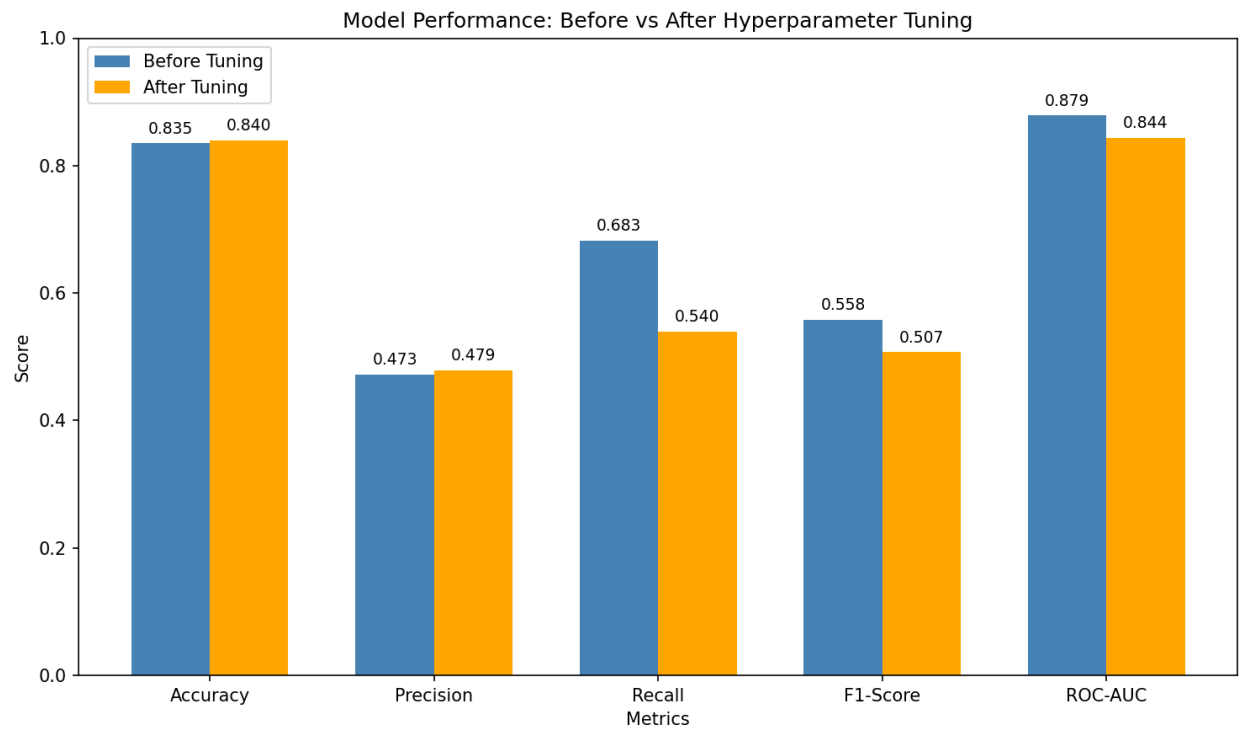


Figure 16: Model performance before vs after hyperparameter tuning

Result: Default Random Forest outperformed the tuned model — original parameters were already near-optimal for this dataset.

6. Business Impact Analysis

6.1 Targeting Strategy Performance

Target Top %	Customers	Response Rate	Lift vs Random Targeting
10%	41	68.3%	+347%
20%	82	50.0%	+227%
30%	123	41.5%	+171%
40%	164	33.5%	+119%
50%	206	29.1%	+90%

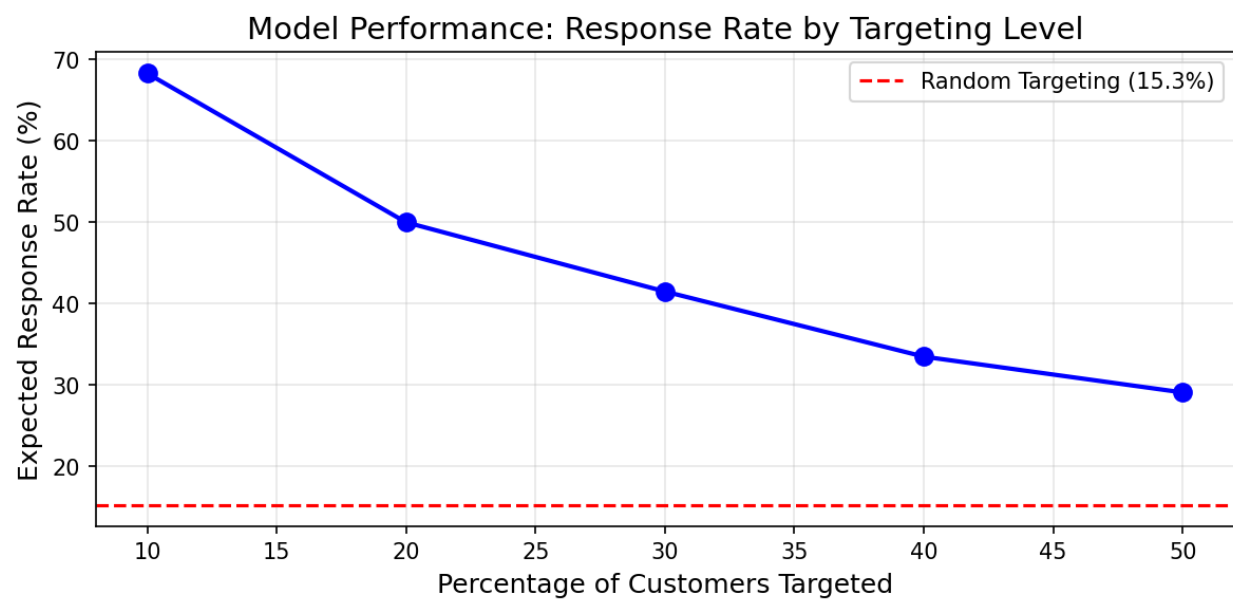


Figure 17: Response rate by targeting level - 227% lift at top 20%

6.2 ROI Calculation

Strategy	Customers	Cost (\$2.50/ea)	Revenue (\$25/response)	Profit	ROI
Target Everyone	412	\$1,030	\$1,575	\$545	53%
Target Top 20% ✓	82	\$205	\$1,025	\$820	300%

Budget Savings \$825 saved (80% cost reduction)	Profit Improvement +\$275 (50% higher profit)
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7. Recommendations

7.1 Immediate Targeting Actions

Priority	Segment	Strategy	Expected Response
1	Ultra Premium	Exclusive offers, VIP treatment, early access	40%+
2	Premium Spenders	Standard campaign inclusion, loyalty rewards	15%
3	Budget Conscious	Discount-driven emails, value messaging	11%
4	Middle Class	Build engagement first — newsletters, brand content	9%

7.2 Model Deployment Steps

- Score all customers using the trained Random Forest model
- Target the top 20% of customers by predicted response probability
- Expected results: ~50% response rate with 80% budget savings

7.3 Key Insights for Marketing Team

Finding	Recommended Action
Past responders are 3x more likely to respond again	Re-engage past campaign responders first
Recent purchasers respond less to campaigns	Wait 30+ days before re-targeting recent buyers
Meat + Wine buyers are highly engaged customers	Cross-sell complementary products to this group
Catalog shoppers show strong responsiveness	Maintain and invest in catalog marketing channel

8. Conclusion

This project delivered four concrete outcomes for the marketing team:

- Segmented customers into 4 actionable groups based on economic behavior
- Built a Random Forest prediction model achieving 55.8% F1-Score and 87.9% ROC-AUC
- Identified the Ultra Premium segment (12.8% of customers) with a 40.7% response rate
- Developed a targeting strategy achieving 227% lift over random campaign targeting

Bottom Line: By targeting only the top 20% of predicted responders, the company can achieve a 50% response rate while saving 80% of marketing budget and increasing profit by 50%.

9. Appendix

9.1 Files Generated

Type	Files
Data	cleaned_customers.csv, clustered_customers.csv, test_predictions.csv
Tables	10 CSV files in reports/tables/
Figures	15+ PNG files in reports/figures/
Model	random_forest_model.pkl, scaler.pkl

9.2 Tools & Technologies Used

- Python 3.11
- pandas, numpy — Data manipulation and processing
- matplotlib, seaborn — Data visualization
- scikit-learn — Machine learning models and evaluation
- imbalanced-learn — SMOTE for handling class imbalance